

## MECHERIA, Algeria

WMO: 605490

Lat: 33.536N

Lon: 0.242W

Elev: 3855

StdP: 12.76

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>29.6 | (c)<br>32.2 | (d)<br>12.0                   | (e)<br>11.7 | (f)<br>54.5 | (g)<br>16.2 | (h)<br>14.4 | (i)<br>53.2 | (j)<br>34.2           | (k)<br>41.8 | (l)<br>32.0 | (m)<br>45.1 | (n)<br>5.4            | (o)<br>260 | (p)<br>0.559 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 25.3                   | 101.0           | 67.2 | 98.9 | 66.3 | 96.9 | 65.5 | 72.6                | 93.8 | 70.7 | 91.4 | 69.0 | 89.3 | 15.5                 | 240  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.3                            | 111.7 | 84.2 | 64.3 | 104.4 | 80.6 | 62.5 | 97.7 | 79.2 | 39.1          | 94.2 | 37.2 | 92.2 | 35.6 | 89.2 | 80.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 30.0              | 26.0 | 22.8 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |
|                   |      |      | 22.9                       | 75.7 | 6.8                | 3.7 | 18.0                                               | 78.3 | 14.0       | 80.5 | 10.2       | 82.5 | 5.3        | 85.2 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 63.2   | 44.1  | 48.1  | 54.4  | 59.9  | 68.0  | 78.0  | 85.3  | 82.4  | 73.3  | 64.7  | 52.3  | 46.5  |  |  |
|                                                                        | DBStd       | 15.14  | 6.01  | 6.02  | 6.75  | 7.18  | 7.85  | 6.66  | 4.66  | 4.47  | 6.38  | 6.13  | 6.47  | 5.13  |  |  |
|                                                                        | HDD50       | 511    | 199   | 93    | 33    | 7     | 1     | 0     | 0     | 0     | 0     | 0     | 50    | 128   |  |  |
|                                                                        | HDD65       | 2753   | 649   | 473   | 335   | 187   | 62    | 1     | 0     | 0     | 10    | 81    | 382   | 573   |  |  |
|                                                                        | CDD50       | 5317   | 15    | 40    | 168   | 303   | 559   | 840   | 1093  | 1004  | 699   | 457   | 119   | 20    |  |  |
|                                                                        | CDD65       | 2083   | 0     | 0     | 5     | 33    | 155   | 391   | 628   | 539   | 258   | 73    | 1     | 0     |  |  |
|                                                                        | CDH74       | 25509  | 0     | 0     | 60    | 383   | 1660  | 4801  | 8772  | 6782  | 2564  | 484   | 3     | 0     |  |  |
|                                                                        | CDH80       | 13464  | 0     | 0     | 7     | 84    | 632   | 2538  | 5301  | 3729  | 1088  | 85    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 10.7   | 10.5  | 11.0  | 11.0  | 13.0  | 12.1  | 11.1  | 11.6  | 10.3  | 9.8   | 8.6   | 10.3  | 9.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 9.5    | 0.7   | 0.7   | 1.1   | 0.9   | 0.9   | 0.4   | 0.2   | 0.5   | 1.2   | 1.4   | 1.0   | 0.6   |  |  |
|                                                                        | PrecMax     | 16.5   | 2.1   | 1.7   | 5.0   | 2.2   | 2.4   | 2.4   | 0.8   | 1.7   | 3.0   | 6.7   | 6.0   | 1.7   |  |  |
|                                                                        | PrecMin     | 5.8    | 0.0   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.4   | 0.0   | 0.0   | 0.0   |  |  |
|                                                                        | PrecStd     | 2.4    | 0.6   | 0.4   | 1.1   | 0.6   | 0.7   | 0.5   | 0.2   | 0.4   | 0.7   | 1.4   | 1.2   | 0.4   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 66.4  | 70.2  | 80.2  | 86.0  | 94.6  | 101.3 | 105.1 | 102.1 | 97.3  | 85.8  | 73.8  | 66.0  |  |  |
|                                                                        |             | MCWB   | 49.7  | 50.0  | 52.8  | 57.1  | 62.0  | 63.5  | 69.3  | 71.1  | 64.5  | 59.1  | 56.0  | 50.4  |  |  |
|                                                                        | 2%          | DB     | 61.5  | 66.2  | 74.8  | 82.3  | 89.8  | 97.9  | 102.4 | 99.0  | 93.6  | 82.3  | 70.8  | 61.8  |  |  |
|                                                                        |             | MCWB   | 46.8  | 48.4  | 51.5  | 55.6  | 61.1  | 64.3  | 67.6  | 67.9  | 64.4  | 58.0  | 54.7  | 48.2  |  |  |
|                                                                        | 5%          | DB     | 57.7  | 62.6  | 71.2  | 78.6  | 86.3  | 95.3  | 100.5 | 97.0  | 90.0  | 79.1  | 67.6  | 59.0  |  |  |
|                                                                        |             | MCWB   | 45.4  | 47.0  | 50.9  | 55.1  | 60.0  | 64.0  | 66.9  | 66.2  | 63.3  | 57.8  | 52.7  | 47.2  |  |  |
|                                                                        | 10%         | DB     | 54.5  | 59.5  | 67.9  | 74.7  | 82.9  | 93.0  | 98.6  | 94.8  | 86.4  | 76.2  | 64.1  | 56.0  |  |  |
|                                                                        |             | MCWB   | 43.7  | 45.8  | 50.0  | 53.9  | 58.6  | 63.3  | 66.2  | 65.6  | 62.2  | 57.8  | 51.4  | 46.6  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 51.6  | 53.0  | 57.0  | 61.3  | 68.3  | 71.1  | 75.3  | 75.2  | 71.9  | 64.9  | 58.8  | 53.3  |  |  |
|                                                                        |             | MCDB   | 61.1  | 64.5  | 71.4  | 77.7  | 85.5  | 92.7  | 97.8  | 98.1  | 89.6  | 75.2  | 70.2  | 61.5  |  |  |
|                                                                        | 2%          | WB     | 49.0  | 50.6  | 54.4  | 58.9  | 65.1  | 68.6  | 73.0  | 72.5  | 68.4  | 62.6  | 56.5  | 50.7  |  |  |
|                                                                        |             | MCDB   | 58.8  | 62.4  | 68.4  | 74.5  | 82.4  | 90.7  | 95.6  | 92.5  | 84.8  | 74.7  | 67.0  | 58.2  |  |  |
|                                                                        | 5%          | WB     | 46.8  | 48.7  | 52.7  | 57.0  | 62.6  | 66.8  | 70.9  | 70.8  | 66.2  | 61.0  | 54.6  | 49.1  |  |  |
|                                                                        |             | MCDB   | 55.0  | 59.5  | 67.0  | 71.6  | 80.3  | 87.9  | 93.8  | 89.8  | 81.4  | 72.2  | 65.0  | 56.0  |  |  |
|                                                                        | 10%         | WB     | 45.0  | 46.9  | 51.1  | 55.4  | 60.6  | 65.1  | 69.1  | 69.0  | 64.5  | 59.6  | 52.6  | 47.6  |  |  |
|                                                                        |             | MCDB   | 52.4  | 57.2  | 65.0  | 69.8  | 77.8  | 85.9  | 92.0  | 86.9  | 80.0  | 70.9  | 61.9  | 54.5  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 16.4  | 18.4  | 20.7  | 21.4  | 22.4  | 23.7  | 25.3  | 23.6  | 21.7  | 19.1  | 16.8  | 16.8  |  |  |
|                                                                        |             | MCDBR  | 21.9  | 23.8  | 25.9  | 26.1  | 25.5  | 25.9  | 27.2  | 25.9  | 24.8  | 22.1  | 20.8  | 22.2  |  |  |
|                                                                        |             | MCWBR  | 12.0  | 11.8  | 11.1  | 10.3  | 10.2  | 9.8   | 10.2  | 9.8   | 9.5   | 8.7   | 10.3  | 11.9  |  |  |
|                                                                        | 5% WB       | MCDBR  | 19.6  | 20.5  | 23.5  | 24.4  | 24.5  | 26.1  | 28.0  | 25.6  | 22.7  | 19.6  | 19.8  | 18.8  |  |  |
|                                                                        |             | MCWBR  | 11.8  | 11.3  | 11.5  | 10.5  | 10.7  | 11.0  | 12.0  | 11.1  | 10.7  | 8.9   | 10.8  | 11.2  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.264 | 0.293 | 0.362 | 0.388 | 0.431 | 0.464 | 0.532 | 0.490 | 0.443 | 0.381 | 0.315 | 0.275 |  |  |
|                                                                        | taud        |        | 2.467 | 2.380 | 2.170 | 2.112 | 2.011 | 1.918 | 1.758 | 1.866 | 1.992 | 2.162 | 2.368 | 2.460 |  |  |
|                                                                        | Ebn at Noon |        | 306   | 305   | 288   | 285   | 272   | 262   | 244   | 252   | 259   | 267   | 281   | 294   |  |  |
|                                                                        | Edh at Noon |        | 27    | 33    | 45    | 50    | 56    | 61    | 71    | 63    | 53    | 41    | 30    | 26    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1061  | 1359  | 1754  | 2095  | 2317  | 2507  | 2420  | 2200  | 1806  | 1461  | 1106  | 956   |  |  |
|                                                                        | RadStd      |        | 77    | 78    | 75    | 100   | 96    | 70    | 64    | 76    | 108   | 95    | 72    | 57    |  |  |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (4 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page